

The Cryptographer Who Outsmarted Himself



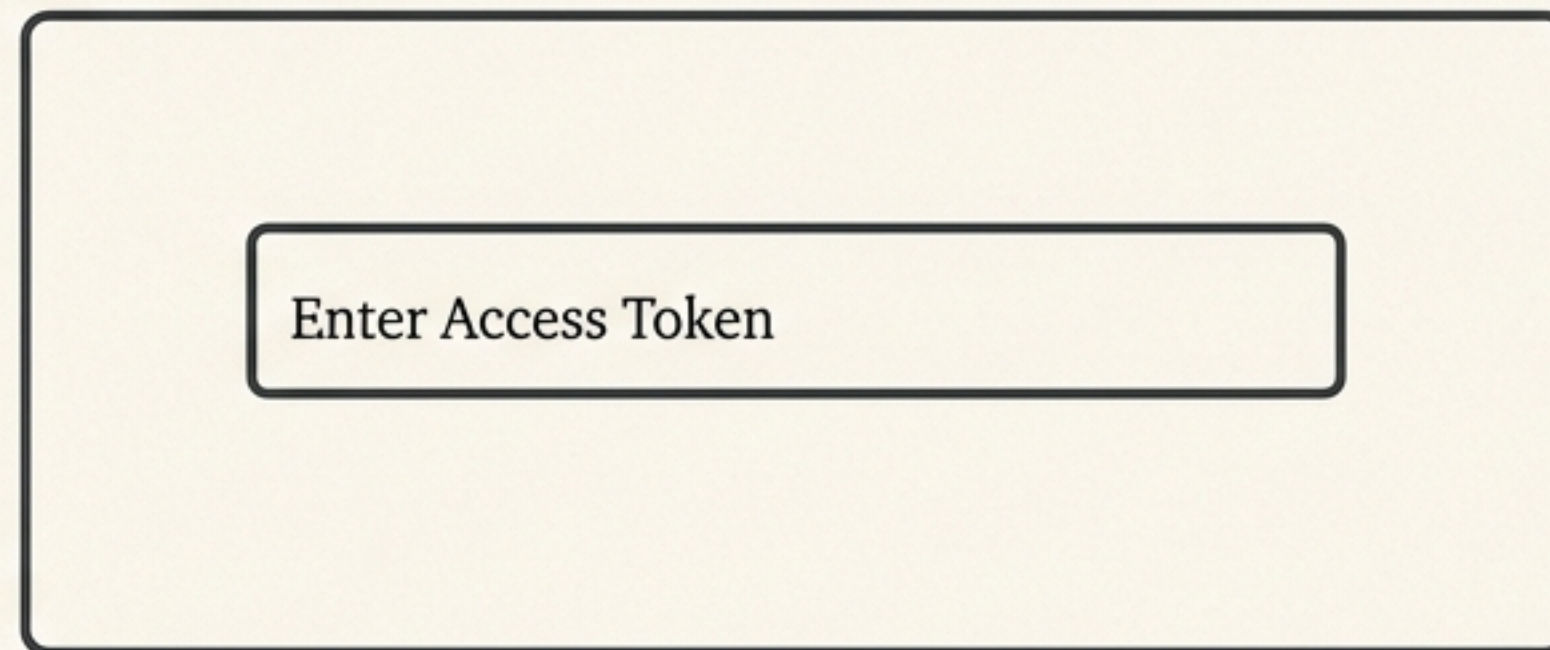
A Field Report from the SGraph
Send Beta Trenches (v0.13.0)

A Perfectly Logical Assumption

Browser A



Browser B



1. Pedro successfully sends a file in Browser A.
2. Opens Browser B. The app asks for an access token.
3. The token is missing.
4. The logical conclusion: "The download URL. The token must be in there somewhere."

An Encryption Key Goes on a Journey

https://secure-file-share.app/download/private#token=eyJhbGciOiJIc2E6IiwiaWF0IjpmZDQzZfYyYzU1MCIslmV4cCI6ImTcwMiQ47h1n-m4_o2_p1_q3_r9_s6_t8

User Expectation: Access Token

Reality: 32 bytes of raw cryptographic entropy

Spec: AES-256-GCM encryption key, base64-encoded to safely live in a URL fragment.

eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9=eyJhbGciOiJI
sinRScCl6IkpXVCJ9.eyJhY2Nlc3NfdG4lOi40MyIuNGFJZS
MGNILTA0ZDQ3ZfyYZU1MCIsImV4ccloMIQ4evliiN3Tj-
ngNILTA0ZDQ3ZfyyYZU1MCIsImVTcwMIQ4=yilb3x5_j-
7h1n-m4_o2_p1_q3_r9_s6_t8

Decoding Chaos

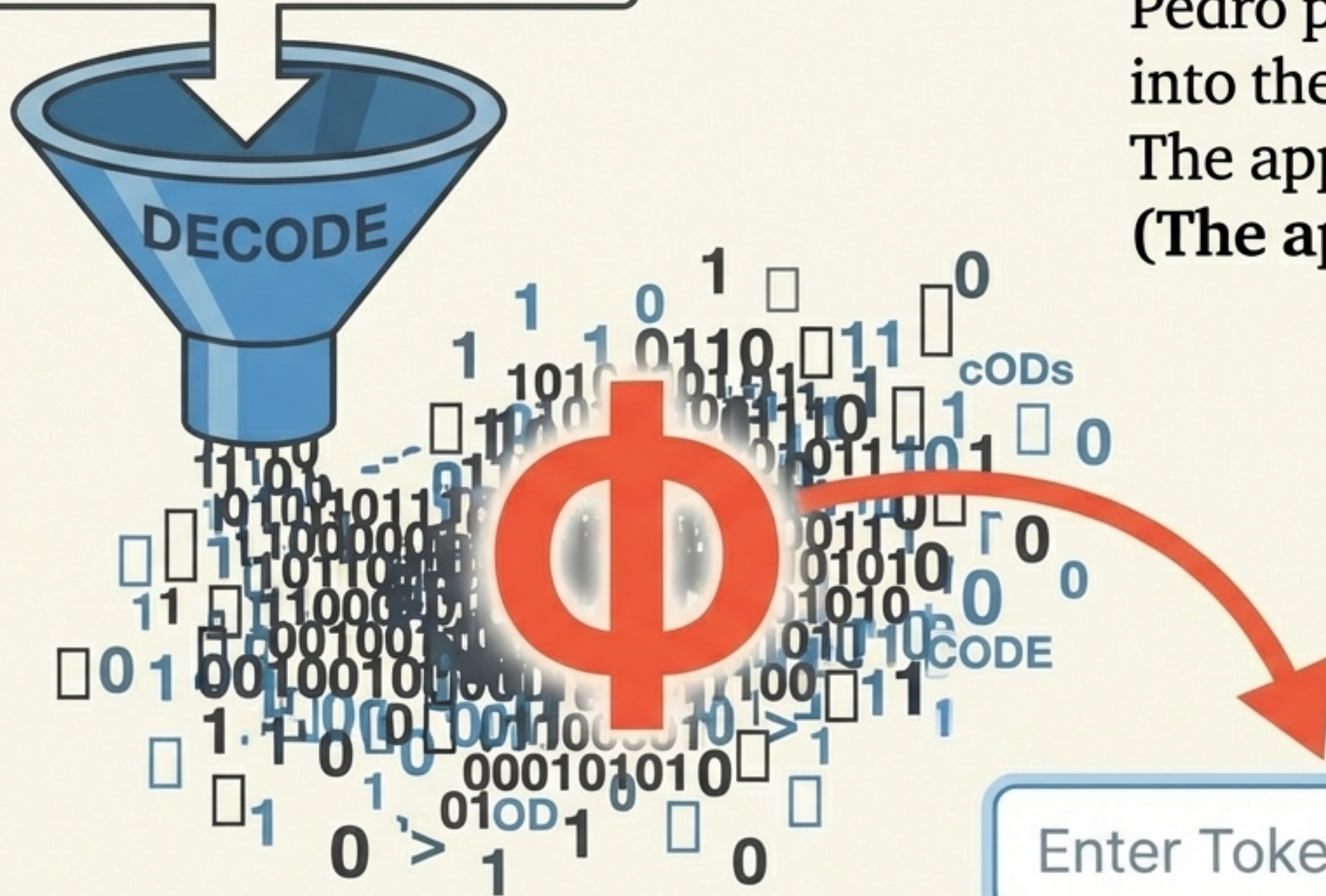
Pedro pastes this binary output into the token input field.

The app says, “Looks good.”

(The app was wrong).

**Unicode character U+0278
(Latin Small Letter 'Phi' (Latin
Small Letter Phi With Stroke))**

Enter Token Here Φ |



GT America Extended: A 1999 Standard Strikes Back

The Raw Error:

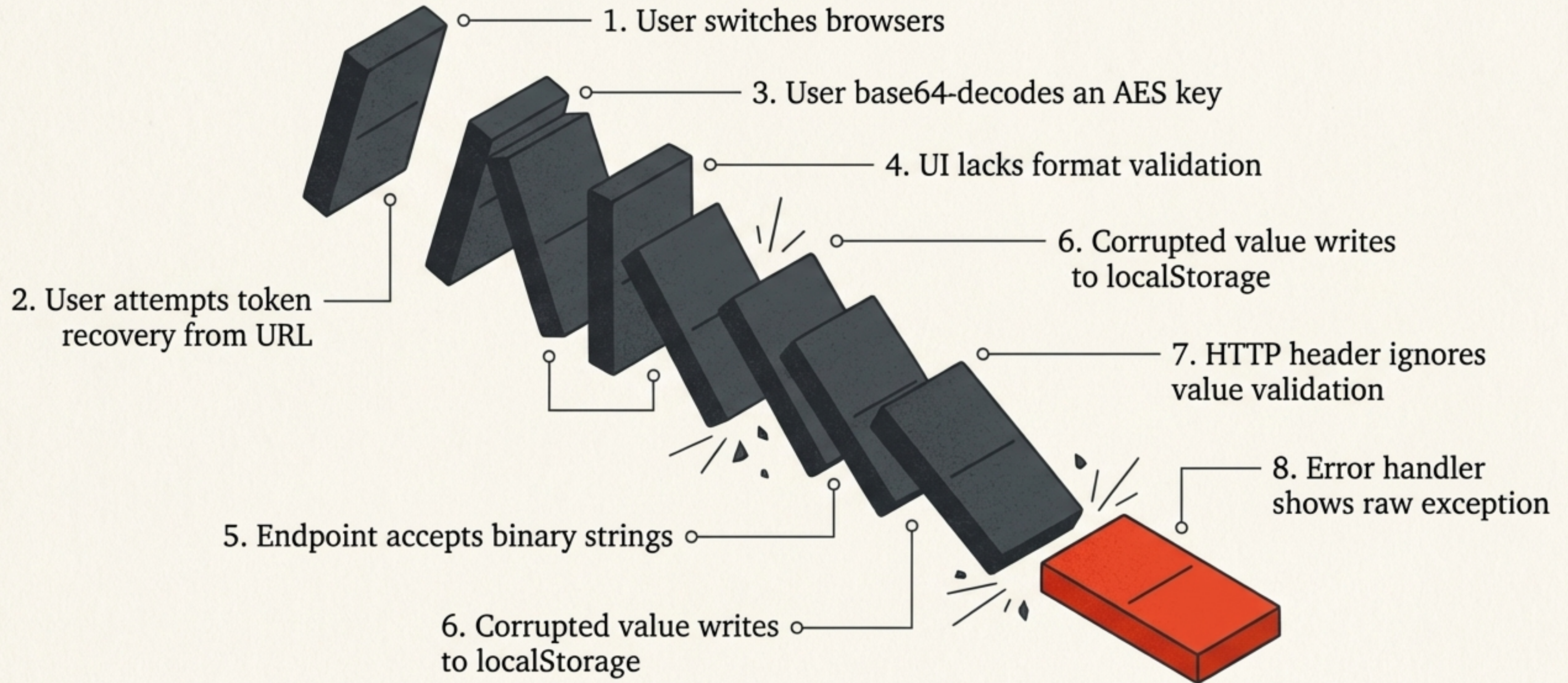
Developer Tools > Console

Failed to execute 'fetch' on 'Window': Failed to read the 'headers' property from 'RequestInit': String contains non ISO-8859-1 code point.

The Impact:

Every upload fails. No explanation. No recovery button. The corrupted token sits in localStorage, detonating on every page load.

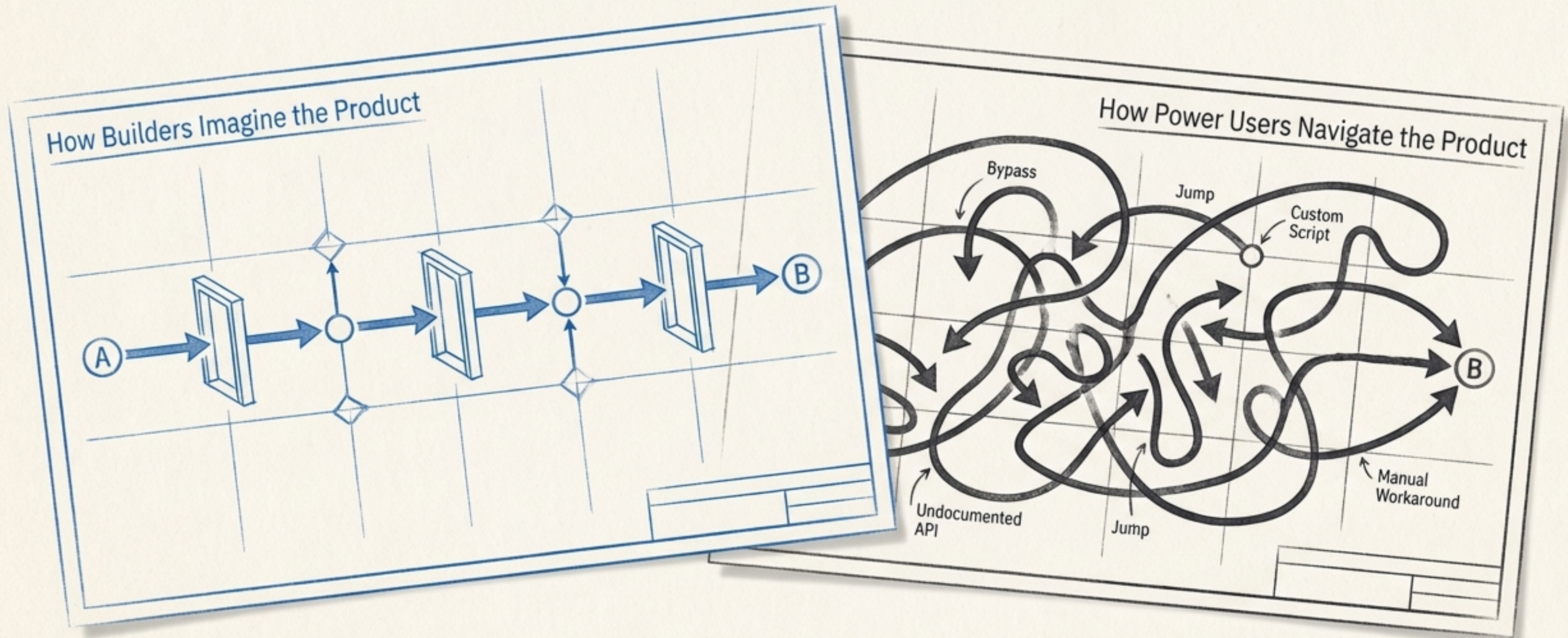
Eight Falling Dominoes



INVESTIGATIVE EDITORIAL

Power Users Have Their Own Mental Models

Technical users find paths you didn't know existed.
They aren't trying to break things. They are trying to make things work.



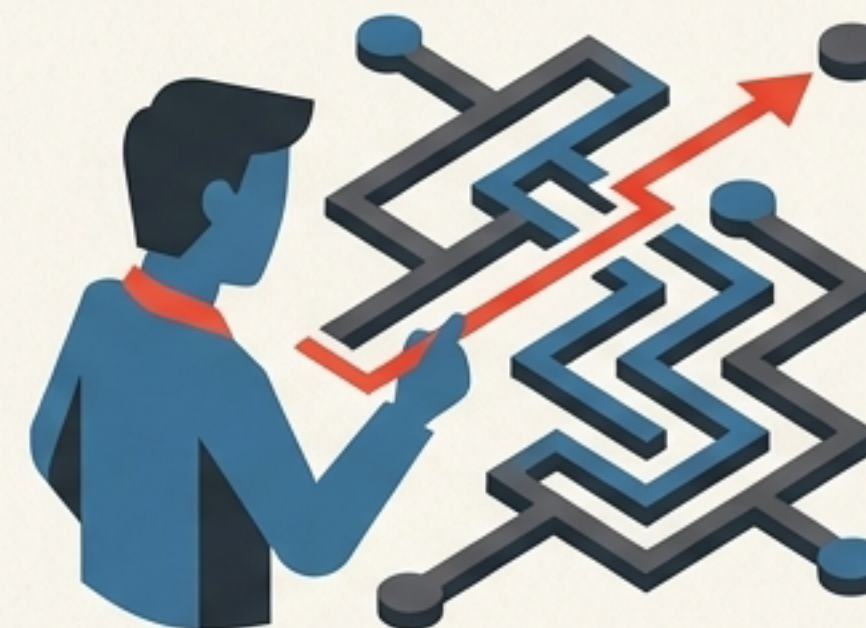
A Taxonomy of Power Users: The Creative Breakers



The Reverse Engineer

“I can figure out how this works by looking at what it produces.”

Decodes URLs, inspects network traffic, pastes binary into text fields.



The Workflow Inventor

“Your UI has 5 steps but I only need 2.”

Bookmarks deep links, shares URLs through unintended channels, composes API endpoints into shell scripts

A Taxonomy of Power Users: The Boundary Pushers



The Edge Case Artist

Brain pattern-matches on boundaries.
Uploads 0-byte files, 4GB files, or
files named ../../../../etc/passwd.



The Concurrency Explorer

Opens three tabs to upload
simultaneously. Uncovers session
conflicts and race conditions.



The Helpful Documenter

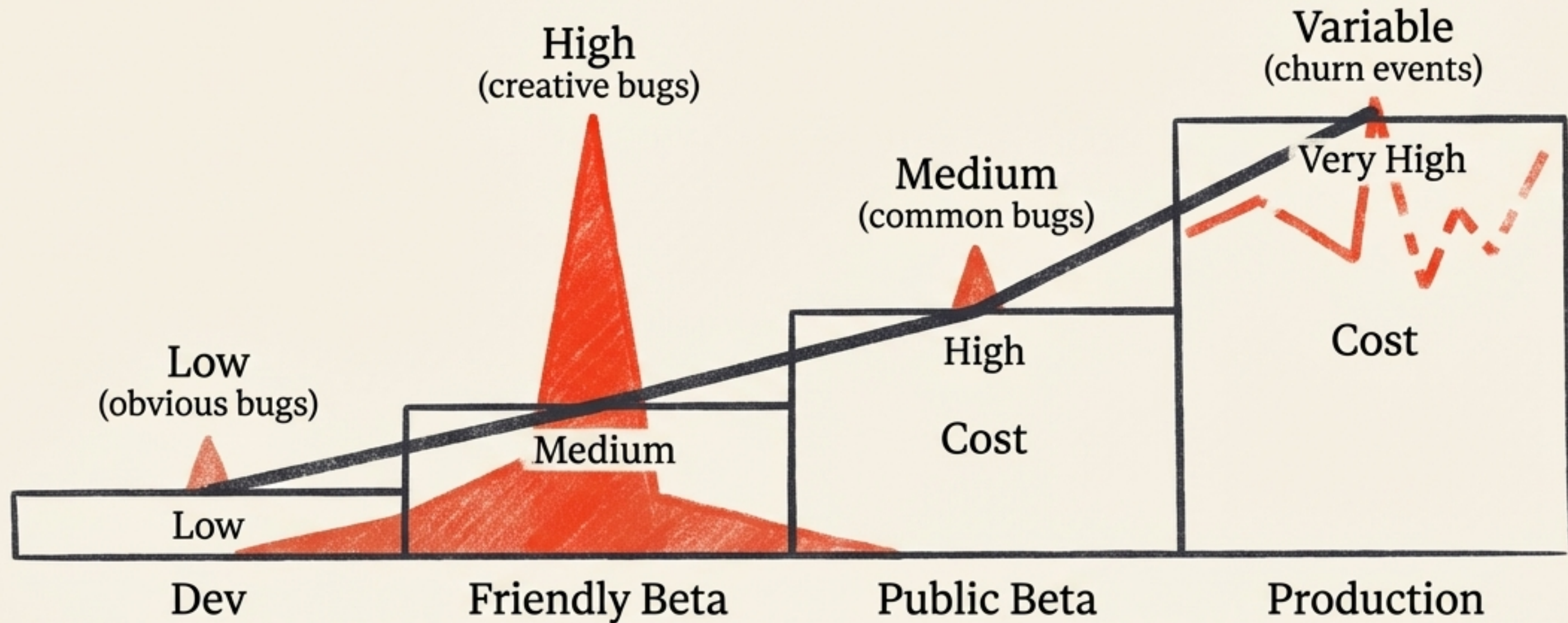
Provides stack traces, network logs,
OS versions, and suggested fixes
alongside their bug reports. A
genuine gift to the team.

Why MVPs Need Friendlies



The ideal beta tester is a ‘Friendly.’ They are technically capable, genuinely invested in your success, and willing to report what they actually did rather than what they think you want to hear.

The Economics of MVP Testing

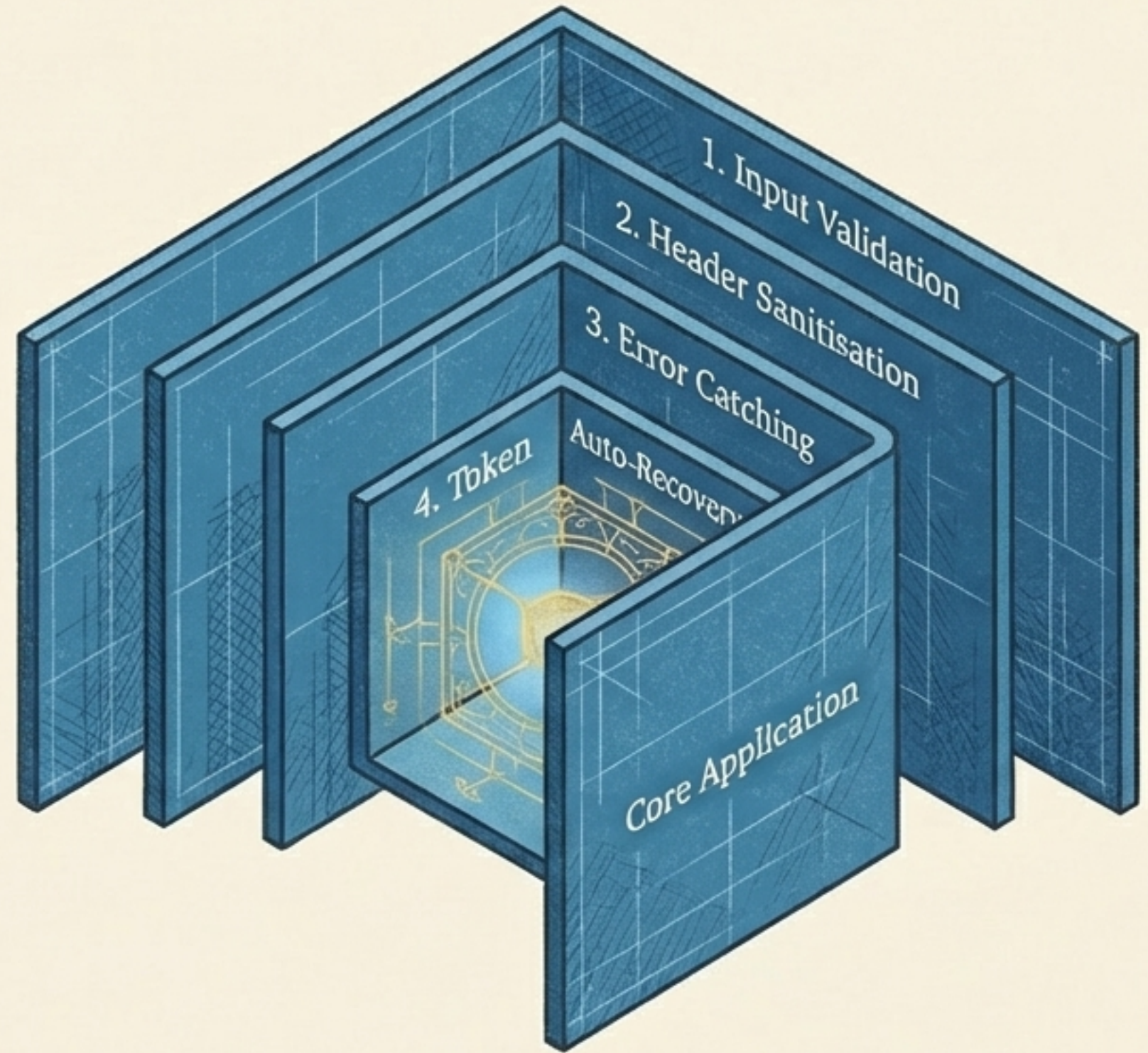


“The Pedro bug cost an afternoon in friendly beta. In production, it would have been an unrecoverable incident.”

Defense in Depth

The single incident revealed a class of vulnerability requiring four structural fixes across three files:

1. Input Validation
2. Header Sanitisation
3. Error Catching
4. Token Auto-Recovery



The Eighteen-Word UX Fix

Access Token

Submit

"Your token is saved in this browser only.
If you switch browsers or devices, you'll
need to enter it again."

If Pedro had seen this hint, the AES key would have stayed encoded, the U+0278 character would have remained theoretical, and the dominoes would never have fallen.

Finding the Gaps in Imagination

Testers don't just find bugs. They find the gaps between what builders imagine and what users actually do. They reveal where a single sentence of explanation prevents a bug from ever being findable.

